Modernizing 3D Graphics Rendering for Interactive Taal Volcano Plume Simulation

A Thesis Proposal

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by

BERNANDINO, Adrian Rafael
BORROMEO, Anton Miguel
GALAN, Russell Emmanuel
GANAYO, Reinman Geoffrey

Neil Patrick Del Gallego, Ph.D.
Adviser

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Abstract

The Taal Volcano in the Philippines remains one of the country's most active volcanoes due to its recurring eruptive activity, highlighting the continued need for effective tools that support hazard communication, disaster preparedness, and scientific visualization. Interactive three-dimensional computer graphics simulations have demonstrated their potential to provide intuitive visual representations of volcanic eruption plumes while allowing users to explore eruption scenarios in an immersive virtual environment. Existing graphics-based volcanic plume simulators have successfully demonstrated real-time visualization capabilities; however, many rely on legacy graphics rendering technologies and software architectures that limit extensibility, maintainability, and the efficient utilization of modern graphics hardware. This study proposes a modern 3D graphics rendering framework for interactive Taal Volcano plume simulation by redesigning the rendering architecture of an existing plume visualization system while preserving its established particle-based simulation model and core visualization techniques. The proposed framework aims to modernize the rendering pipeline, graphics resource management, shader implementation, and software architecture to provide a more scalable, maintainable, and extensible platform for future graphics-based scientific visualization applications. A design and development research methodology will be employed to reimplement the simulator using a contemporary graphics rendering framework and evaluate its rendering performance, graphics resource utilization, scalability, and visual fidelity through quantitative performance metrics and qualitative visual assessment.

Keywords: Computer graphics, 3D graphics, Graphics rendering, Volcanic plume simulation, Taal Volcano, Particle systems

Contents

Chapter 1

Introduction

1.1 Background of the Study

Volcanic eruptions are among the most destructive natural hazards, producing ash plumes, pyroclastic flows, and volcanic gases that threaten nearby communities and critical infrastructure. Effective visualization of these hazards plays an important role in disaster preparedness, scientific communication, and public education. Traditionally, volcanic simulations have relied on numerical models that prioritize scientific accuracy for hazard prediction and analysis. While these models provide valuable quantitative information, they are often computationally intensive and lack the interactive capabilities necessary for real-time visualization (Rossant & Rougier, 2021).

To overcome these limitations, the subfield of real-time computer graphics has increasingly been applied to scientific visualization, balancing visual plausibility with high interactive performance. Within this domain, a notable localized development is AnitoPlume, a real-time three-dimensional volcanic plume simulator designed specifically for visualizing the complex eruption dynamics of Taal Volcano (Del Gallego et al., 2026).

This specialized application combines a Lagrangian particle-based plume system with a detailed digital elevation model of the volcano's terrain. By maintaining an interactive rendering performance of approximately 30 frames per second, the system demonstrated that computer graphics could bridge the gap between complex geological events and accessible hazard-awareness software for local education and planning (Del Gallego et al., 2026).

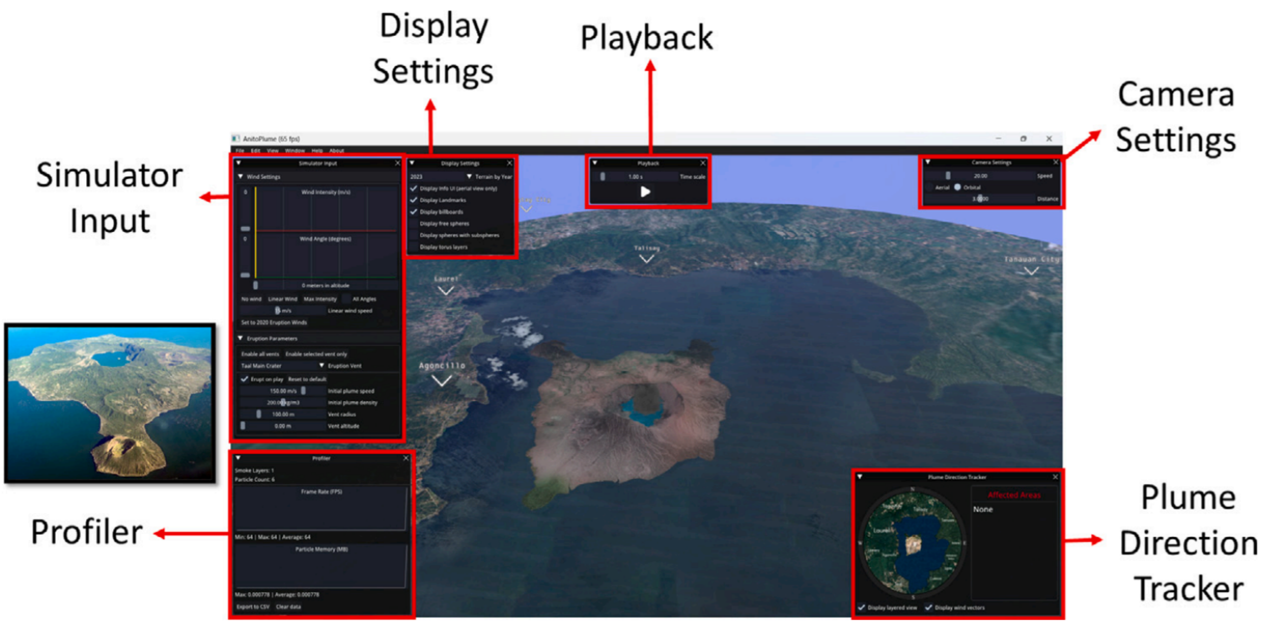

Figure 1: User Interface of AnitoPlume. The Taal volcano model is rendered in the center of the simulator. The photo reference is on the left. (Del Gallego et al., 2026).

Previous efforts in graphics-driven volcanic visualization have heavily relied on traditional, high-level graphics architectures due to their broad portability and structural simplicity (Shiraef, 2016). In these legacy state-machine models, the graphics API simplifies software development by delegating critical hardware management tasks, such as memory allocation and command scheduling, directly to the vendor's graphics driver. However, as visualization workloads scale, this abstraction inevitably introduces massive CPU driver overhead and restricts developers from optimizing how the hardware executes commands (Bossard, 2019). Consequently, the computer graphics industry has seen a global shift toward modern, explicit rendering paradigms. These low-level architectures strip away hidden driver intervention and grant developers granular, direct control over GPU resources, pipeline states, and memory synchronization to maximize hardware efficiency (Ioannidis & Boutsi, 2020a).

Despite the theoretical performance advantages of modern explicit rendering pipelines, a significant challenge remains: the architectural enhancement of established scientific tools. High-level visualization platforms like the original AnitoPlume simulator are tightly coupled to legacy architectures, creating a rigid barrier to optimization and preventing the integration of advanced, contemporary rendering techniques. Transitioning an active simulation system to an explicit

Figure 2: Graphics APIs. (TechArx Gaming Staff, 2016).

framework requires a complete restructuring of shader programs, command submissions, and memory allocation strategies. This process moves beyond a mere software translation; it presents a critical opportunity to upgrade the shading architecture and optimize the visual representation of complex fluid-terrain interactions (Unterguggenberger, Kerbl, & Wimmer, 2022). Therefore, investigating how explicit GPU control impacts rendering throughput, memory bandwidth utilization, and overall visual fidelity provides essential insights for the future development of highly scalable scientific visualization systems.

1.2 Research Objectives

The primary aim of this study is to formulate and evaluate a low-overhead, explicit rendering architecture for the AnitoPlume volcanic plume simulator, and to quantify the impact of explicit GPU control on volumetric rendering throughput, memory bandwidth utilization, and visual fidelity.

The specific objectives of this research are as follows:

1. To design an explicit volumetric rendering pipeline that transitions the AnitoPlume simulator from a dynamic state-machine model to explicitly defined Pipeline State Objects (PSOs) and command lists, while strictly preserving the baseline physical simulation accuracy;

2. To implement and assess explicit GPU resource synchronization and cross-API shader translation pipelines to manage the structural transition and

memory allocation of large volumetric datasets; and

3. To analyze the rendering efficiency of the modernized pipeline against the previous implementation, utilizing GPU profiling metrics, including frame pacing, shader execution time, memory bandwidth utilization, and overall render throughput under identical hardware constraints.

4. To verify that the architectural design maintains or improves the interactive user experience by conducting usability testing with target beneficiaries.

1.3 Scope and Limitations of the Research

The implementation evaluates an explicit graphics architecture using the existing AnitoPlume visualization framework as the simulation platform. The study investigates explicit pipeline state management, command recording, shader portability, and resource synchronization while preserving the underlying plume simulation model. Modifications to the scientific simulation algorithms are outside the scope of this work.

1.4 Significance of the Research

This study contributes to both computer graphics research and scientific visualization by providing empirical evidence on the application of explicit graphics architectures to real-time volumetric scientific visualization. For researchers in computer graphics, the study offers a detailed case study demonstrating the architectural modifications required when transitioning between two fundamentally different graphics APIs. The comparison between the use of an implicit to an explicit graphics API architecture contributes to empirical evidence regarding rendering performance, implementation complexity, and software maintainability.

For developers of scientific visualization software, the findings may serve as a reference for future projects that require designing future high-performance rendering systems. For future researchers, this work establishes a scalable, explicit rendering foundation that supports integrating advanced graphics techniques, including hardware-accelerated ray tracing, compute shaders, GPU-driven rendering, and other contemporary visualization methods. Consequently, the study extends the long-term applicability of the original AnitoPlume project beyond its initial legacy implementation.

Chapter 2

Related Works

2.1 Taal Volcano Primer

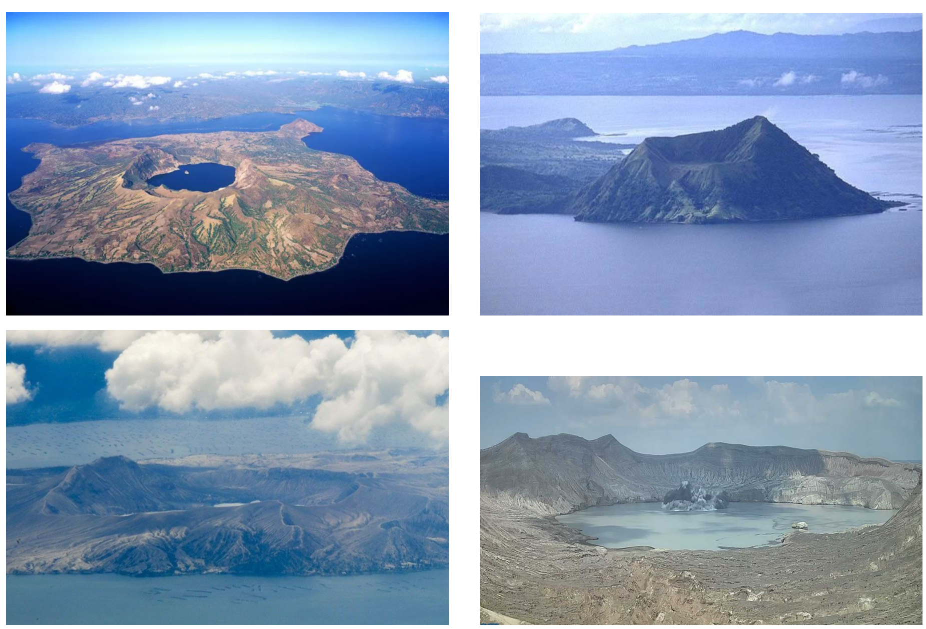

Figure 3: From left to right, top to bottom: Taal volcano during 2015, 2019, 2023, and 2026 (TechArx Gaming Staff, 2016; Flores, 2019; Daguno-Bersamina, 2023; Mallari Jr., 2026).

Taal Volcano is a complex volcano located in southwestern Luzon, Philippines, situated in the middle of Taal Lake (Smithsonian Institution, 2026). Figure 3 illustrates the volcano across different time periods, including its primary eruption phases. As one of the most active volcanoes in the country, the Philippine Institute of Volcanology and Seismology (PHIVOLCS) has recorded 36 historical eruptions at Taal (PHIVOLCS, 2024). Its frequent activity poses a major threat to local communities and the densely populated Metro Manila area. Although official hazard maps designate only the volcano island as a Permanent Danger Zone (PDZ) and outline a high-risk perimeter of 15 km, Taal's actual danger extends much further due to the regular release of toxic gases. Consequently, Metro Manila remains highly vulnerable to these emissions despite being located approximately 81 km away from the volcanic center.

This vulnerability was clearly demonstrated during the January 12, 2020 eruption, when a phreatomagmatic event occurred at the main crater, producing a massive column of ash and steam. The severity of the eruption led PHIVOLCS to raise the alert status to Level 4, indicating that a hazardous eruption was imminent. The event was classified as a Volcanic Explosivity Index (VEI) 4 eruption, depositing an estimated tephra fallout volume of 0.097 km^3 (Balangue-Tarriela et al., 2022). Due to the immediate risks, a mass evacuation of roughly 376,000 people was carried out in the surrounding areas, including the provinces of Batangas and Cavite (Renzo, 2020; Del Castillo, Paraiso, Vicente, Jamero, & Narisma, 2020).

The 2020 event is the only major eruption of Taal to be thoroughly documented, thanks to modern communication and technologies like accessible digital cameras, crowdsourcing, and high-resolution seismic sensors. This availability of real eruption footage and geographical data provides a strong foundation for developing visual simulations of the volcano.

2.2 Geoscientific Modeling and Volcanic Crisis Communication

2.2.1 Limitations of Current Numerical Models

In volcanology, the standard approach for evaluating risk and hazard zones relies on numerical simulations that model the physics of ash transport and tephra fallout. Volcanologists widely use Eulerian models such as FALL3D (Folch, Costa, & Macedonio, 2009), Tephra2 (Bonadonna, 2006; Connor & Connor, 2006), and TephraProb (Biass, Bonadonna, Connor, & Connor, 2016), alongside Lagrangian models like HYSPLIT (Stein et al., 2015), for scientific analysis and hazard mapping. These models are highly valued for their quantitative accuracy and their capacity to incorporate atmospheric variables.

However, these simulations are built for expert analysis rather than public situational awareness, requiring high-performance computing and substantial processing time. While these methods are essential for long-term hazard forecasting, they lack the interactive flexibility and 3D immersion needed for real-time visualization applications. To address this, our proposed software, AnitoPlume, serves as a computer graphics application that complements these numerical models by offering an interactive testbed for 3D volcanic eruption visualizations of Taal Volcano.

2.2.2 Advancements in 3D Hazard Visualization

Traditionally, volcanic hazards and crisis information are communicated using 2D maps that display contour lines and heat maps for ash distribution, pyroclastic flow boundaries, and tephra thickness (Nave, Isaia, Vilardo, & Barclay, 2010; Thompson, Lindsay, & Gaillard, 2015; Fearnley & Beaven, 2018; Esposti Ongaro, Cerminara, Charbonnier, Lube, & Valentine, 2020). Although 2D representations remain the standard for scientific experts, research indicates that integrating "spatial familiarity" is critical for prompting effective public responses (Preppernau & Jenny, 2015). Comparative studies further demonstrate that general users often prefer 3D perspectives over 2D maps, as 3D views make it easier to understand complex terrain features and identify viable evacuation routes (Thompson, Lindsay, & Leonard, 2017; Driedger et al., 2024).

To improve public communication, several research initiatives have explored immersive and interactive applications using computer graphics (Zhang, Kong, Li, Wang, & Qin, 2017; Lastic et al., 2022; Pretorius et al., 2024), virtual reality (VR) (Tibaldi et al., 2020; Asgary, Bonadonna, & Frischknecht, 2019; Li, 2023), or gamification (Kerlow, Pedreros, & Albert, 2020). Tools such as VolcanoVR (Hansen et al., 2025), VRVOLC (Delage et al., 2021), and HoloVulcano (Asgary et al., 2019) utilize commercial game engines to let stakeholders view complex volcanic phenomena in safe, immersive environments.

Distinct from these commercial engine approaches, the original AnitoPlume framework was developed using a custom, open-source stack (C++, OpenGL, Eigen, dearIMGUI) to bypass general-purpose engine abstractions and optimize real-time rendering performance. Furthermore, while generic eruption simulations exist (Lastic et al., 2022; Pretorius et al., 2024), AnitoPlume specifically targeted the high-risk Taal "Decade Volcano" by integrating LiDAR and OpenTopography terrain datasets with real-time plume dynamics. Building directly upon this specific, site-tailored approach, our work modernizes this legacy architecture by replacing its original rendering backend with a next-generation graphics API, significantly enhancing visual fidelity and pipeline efficiency.

2.3 Geoscientific Modeling and Volcanic Crisis Communication

Volcanic modeling paradigms in geosciences and computer graphics differ significantly in both methodology and primary intent. Within geoscientific and volcanological research, the priority is predictive accuracy and scientific forecasting. In this context, numerical solvers are essential for quantifying physical parameters such as mass eruption rates and tephra dispersal (Kavanagh, Engwell, & Martin, 2018; Poland & Anderson, 2020; Esposti Ongaro et al., 2020).

Conversely, in the field of computer graphics, volcanic modeling is treated as a subfield of real-time rendering and procedural simulation, focusing primarily on particle systems and fluid dynamics (Stora, Agliati, Cani, Neyret, & Gascuel, 1999; Roth & Guritz, 1995; Lastic et al., 2022; Li, 2023; Pretorius et al., 2024). Research in this domain emphasizes real-time user interactivity and immersion, meaning that visual effects are often approximations and physical models are simplified. Consequently, these techniques prioritize visual realism and low-latency performance over the complex computations found in traditional numerical solvers.

2.4 AnitoPlume

AnitoPlume is the first interactive, real-time 3D graphics-based simulation application explicitly tailored for visualizing the changing landscape of the Taal Volcano island in the Philippines and its corresponding eruption plumes. Developed primarily to address the challenges of volcanic hazard and crisis communication, the simulator serves as an immersive, lightweight alternative to traditional 2D heatmaps and highly complex, non-interactive geoscientific numerical solvers (such as FALL3D and Tephra2). While geoscientific solvers excel at quantitative predictive accuracy, they require heavy high-performance computing resources and significant processing time, making them unviable for applications requiring immediate visual feedback, interactive flexibility, and low-latency user immersion. AnitoPlume fills this gap by merging computer graphics techniques with site-specific real-world terrain details to facilitate risk awareness and "what-if" situational modeling (Del Gallego et al., 2026). To visually demonstrate these differences in rendering approach and graphical fidelity, Table 1 compares the output of previous plume simulations with the real-time terrain integration of our proposed simulator.

	Geoscientific Models (e.g., FALL3D, Tephra2)	**Computer Graphics Related Work (e.g., Lastic et al. (2022))**	**AnitoPlume (Proposed Simulator)**
Method	Numerical solvers, Eulerian atmospheric dispersion models, Tephra accumulation and dispersion.	Eulerian, Lagrangian, or hybrid particle systems approximation.	Lagrangian system with synchronized multi-vent plume logic.
Visualization	Static 2D/3D low-fidelity contours, graph-based and/or scientific visualization, heatmaps.	3D scene using simple spheres, billboard particles, static planes, or basic landscapes.	3D scene using LiDAR and OpenTopography terrain data, dynamic textures, textured particles, interactive UI with plume trackers.
Performance	High accuracy at the cost of high computation time.	Real-time performance prioritized for visual realism. Approximation of physical eruption behavior.	Real-time performance with improved visual realism. Optimized for 30 FPS. E.g., Terrain-specific model through multi-source 3D modeling workflow.
Temporal Representation	Focuses on snapshots of single eruptive events.	Continuous loop simulations with emphasis on plume structure alone.	Dynamic textures: Multi-year landscape visualization (2015–2023), different plume particles based on duration.
User Interface (UI)	Displays comprehensive outputs and calculations. E.g., dispersal, mass accumulation, probability maps.	Parameter tuning for visual and physics adjustments, either via the UI or via text/command-line input.	A user application requiring no text prompt. Revised layout for parameter tuning. UI is suited for "what-if" scenarios because of integrated hazard trackers.
Validation Strategy	Field-constrained analysis of tephra fall deposits.	Qualitative visual comparisons	Quantitative similarity analysis — structural image similarity (SSIM) and root mean squared error (RMSE). Usability testing conducted.

Table 1: Comparison of previous plume simulation (left) implement by Lastic et al. (2022) and geoscientific numerical solvers (middle) with the AnitoPlume simulator (right) by Del Gallego et al. (2026).

2.4.1 Graphics Architecture and System Performance

To bypass the overhead, heavy abstractions, and unneeded general-purpose features of commercial game engines, AnitoPlume was custom-built using a lean, native C++ open-source API stack consisting of OpenGL for graphics rendering, Eigen for numerical linear algebra, and dearIMGUI for the user interface layout. This custom application architecture optimizes real-time performance on consumer-grade hardware. By keeping the system dependencies minimal, the simulator successfully achieves real-time interactivity, running at an average rendering rate of 30 frames per second (FPS) on a system with 6 GB of video RAM (VRAM) capped at 2080 x 2080 resolution(Del Gallego et al., 2026).

2.4.2 Usability Design and Interactive Features

The simulator also introduced several usability improvements over previous volcanic visualization systems, including simplified camera controls, interactive parameter editing, real-time plume direction tracking, and contextual information about nearby communities affected by volcanic activity. These additions improved both accessibility and user experience while preserving interactive performance (Del Gallego et al., 2026).

2.4.3 System Validation and Evaluation Metrics

To evaluate the effectiveness of the simulator, Del Gallego et al. conducted quantitative and qualitative assessments. Visual fidelity was measured using the Structural Similarity Index (SSIM) and Root Mean Squared Error (RMSE) by comparing simulated eruption images with real-world photographs and surveillance footage of Taal Volcano. The study reported high visual similarity, indicating that the synthesized plume closely resembled actual volcanic eruptions. In addition, usability testing using the System Usability Scale (SUS) demonstrated that participants generally found the simulator easy to use and suitable for interactive exploration.

2.4.4 Architectural Limitations and Future Trajectories

Despite these accomplishments, the primary contribution of AnitoPlume lies in the design of its visualization framework rather than the underlying graphics API. The study focused on plume simulation, terrain modeling, visual fidelity, and usability evaluation, while the OpenGL implementation served primarily as the rendering platform supporting these objectives. Consequently, the original work did not investigate how the rendering architecture itself might influence performance, resource utilization, maintainability, or future extensibility.

The authors also acknowledged several opportunities for future development, which include:

1. Photorealistic Rendering
2. Expanded Physics-Based Mechanics
3. Image-Based Parameter Estimation

4. Geoscience Data Integration
5. Automated Terrain Synthesis

These recommendations highlight the critical need for a modern rendering architecture and a low-level API like DirectX 12, which are capable of supporting such intensive graphics and parallel computing workloads while maintaining real-time interactive performance.

2.5 API Migration in Rendering Systems

The migration of rendering software between graphics APIs is a well-known challenge in computer graphics. It often involves more than just replacing function calls. A successful port must account for differences in shader languages, resource-binding models, render-target handling, memory synchronization, and execution semantics. In practice, this means that a program cannot simply be translated line by line; instead, the rendering architecture must be reinterpreted in the context of the target API.

Previous work in graphics engineering has shown that API migration is most successful when the original system is structured in a modular way. Systems that separate rendering logic, scene management, and platform-specific code are often easier to port than tightly coupled implementations. This is particularly relevant for research software, where maintainability and clarity are essential. The process of porting from OpenGL to DirectX, therefore, becomes not only a technical exercise, but also an opportunity to improve the software architecture.

2.5.1 Cross-API Shader Translation

A major issue in porting visual effects from OpenGL to DirectX 12 is shader translation. OpenGL workflows usually rely on the OpenGL Shading Language (GLSL), while DirectX 12 expects the High-Level Shader Language (HLSL) and a more structured offline compilation pipeline. Existing developer practice tends to fall into two categories: manual shader rewriting for tight control over hardware-specific optimizations, or cross-compilation workflows that preserve shader intent while reducing rewrite cost (Capodieci, Cavicchioli, & Marongiu, 2022).

For a visually sensitive effect such as AnitoPlume's volcanic plumes, the second approach is highly attractive. Manual translation risks introducing numerical drift and subtle visual discrepancies in lighting and blending logic. This design philosophy is exemplified by modern research rendering frameworks like NVIDIA's Falcor, which architecturally separates program descriptions from backend compilation. By utilizing intermediate representations like SPIR-V, such engines prioritize shader portability over hard-coded backend assumptions. In a migration study, this supports the argument that accurate visual porting is not just about translating syntax from GLSL to HLSL, but about maintaining shader behavior across compilation targets using tool-assisted pipelines like the DirectX Shader Compiler (DXC) or SPIRV-Cross (Kinkor, 2022).

2.5.2 Pipeline State Management Adaptation

A second major challenge is the shift from OpenGL's dynamic state model to DirectX 12's explicit Pipeline State Objects (PSOs). In architectures, OpenGL allows rendering states to be changed incrementally at runtime, with the graphics driver silently inferring and tracking state combinations (Shiraef, 2016). DirectX 12 completely eliminates this hidden driver-side management. Instead, explicit APIs require developers to bake the blend state, rasterizer state, depth-stencil state, primitive topology, and shader kernels into immutable PSOs ahead of time (Bossard, 2019).

The necessity of this structural rewrite is evident in advanced rendering architectures like the Falcor framework. Rather than relying on hidden driver-side state inference, Falcor explicitly aggregates vertex layouts, shader kernels, rasterizer states, and framebuffer descriptions into a single, pre-compiled state object definition. At draw time, the render context binds this monolithic object directly before issuing the call. For plume rendering, adopting this explicit design is critical; even minor mismatches in depth testing or alpha blending across different PSOs can fundamentally alter how overlapping particle billboards accumulate and fade (Del Gallego et al., 2026). Therefore, PSO migration must be framed as a foundational architectural overhaul rather than a simple API substitution (Asplund, 2020).

2.5.3 Resource Synchronization and Memory Barriers

The most technically demanding aspect of the explicit migration process is resource synchronization. In OpenGL, hardware hazards are concealed by the driver, meaning developers rarely need to explicitly manage the transitions between writing to and reading from the same texture or buffer (Unterguggenberger et al., 2022). DirectX 12, by contrast, transfers the burden of memory management entirely to the developer, requiring explicit resource barriers to ensure that GPU accesses occur in the correct deterministic order (Rossant & Rougier, 2021).

This explicit synchronization burden is concretely demonstrated in the Falcor engine, which utilizes a dedicated barrier system to route resources into their correct states before any updates, copies, or draw calls are executed. By manually handling transitions at both the resource and subresource levels, the architecture reinforces that state transitions are a central engine responsibility rather than an invisible driver service. For the AnitoPlume study, this supports a key argument: accurate migration is not just about producing a similar output image, but about guaranteeing that the GPU views each buffer in the correct state at the exact right microsecond. This is essential when updating thousands of particle billboards, as incorrect barrier placement during parallel draw calls leads to race conditions, stale texture reads, flickering artifacts, or catastrophic device crashes (Cheung, 2024).

Chapter 3

Theoretical Framework

The modernization of AnitoPlume involves migrating the rendering pipeline from a traditional to a modern graphics API architecture. This migration introduces a fundamentally different graphics architecture characterized by explicit resource management, reduced driver overhead, and improved utilization of modern multi-core processors. These architectural changes are beneficial for simulators such as AnitoPlume, which require continuous updates of thousands of dynamic particle billboards.

3.1 Graphics Rendering Architectures

Traditional graphics API architectures are designed to abstract low-level graphics operations from the programmer by implicitly managing graphics state, resource synchronization, and command validation within the graphics driver. Throughout this study, these are referred to as implicit graphics APIs, whereas architectures that expose these responsibilities directly to the application are referred to as explicit graphics APIs.

3.1.1 Overview of Graphics API Architectures

Implicit graphics APIs such as OpenGL are based on a state-based programming model where the application modifies global rendering states that are maintained by the graphics implementation ("The OpenGL Graphics System: A Specification", 2022). Implicit graphics API architecture relies on a thick driver layer that abstracts graphics pipeline management from the application (Fig. 1). Because the driver is responsible for maintaining graphics state, performing runtime validation, tracking resource usage, and managing command execution, significant CPU overhead is introduced during rendering (Ping et al., 2026).

Explicit graphics architectures expose synchronization, resource management, and command submission responsibilities directly to applications (Khronos Group, 2024). Rather than relying on implicit driver-managed state changes, explicit graphics architectures require the application to explicitly define and manage the configuration of the graphics pipeline. Applications explicitly define pipeline states, resource allocation, synchronization, and memory management as seen on figure 4.

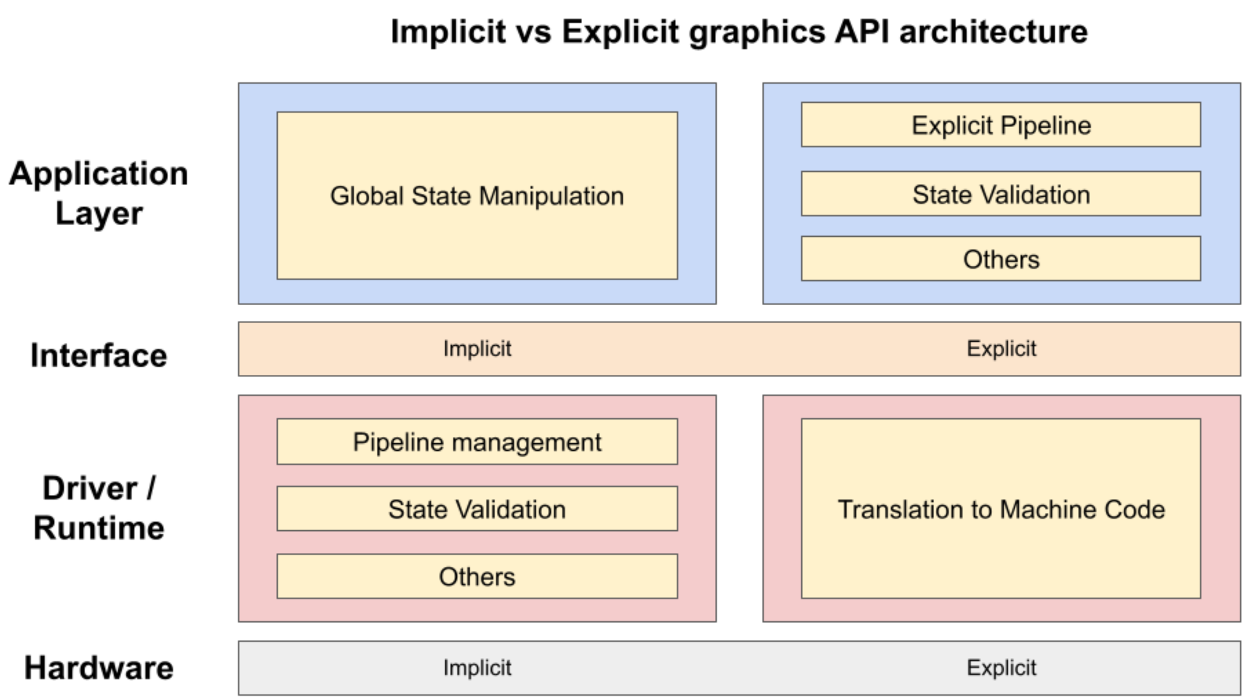

Figure 4: Comparison of implicit graphics API architecture (left) with explicit graphics API architecture (right).

3.1.2 Limitations of the Implicit API Architecture

The limitations of the traditional graphics API architecture become particularly significant in computationally intensive scientific visualization systems such as AnitoPlume. In addition to numerically integrating volcanic plume dynamics including gravity, buoyancy, drag, and wind forces for thousands of particles each simulation step, the CPU must also handle graphics driver validation and state management (Del Gallego et al., 2026). This added overhead limits rendering performance, motivating the use of lower-overhead graphics APIs to reduce CPU-side workload for applications that require high-throughput rendering. As rendering workloads become more complex, these driver-managed operations can limit the application's ability to optimize resource usage, creating CPU-side bottlenecks that reduce the rate at which work can be submitted to the GPU (Ping et al., 2026).

3.1.3 Performance Optimization opportunities

Explicit graphics architectures reduce driver-managed serialization by shifting the responsibility to the application. Tasks such as pipeline configuration, resource allocation, and command preparation can be performed explicitly by the application before or outside the main rendering loop. During runtime, the system only executes predefined instructions without modifying or accessing the underlying data. This shift allows command generation to be safely distributed across multiple CPU threads (Ping et al., 2026). The lightweight driver architecture introduced by explicit graphics APIs minimizes runtime validation and hidden state management, reducing CPU overhead while providing developers with greater control over rendering operations. By enabling application-driven optimization and more efficient utilization of hardware resources, explicit graphics API architectures are better suited for performance-critical rendering workloads (Ioannidis & Boutsi, 2020b).

3.2 Lagrangian Model for Volumetric Plume Dynamics

AnitoPlume utilizes a Lagrangian formulation for representing volcanic plume evolution, where the plume is modeled as a collection of discrete volumetric elements whose physical states are updated over time. Unlike Eulerian approaches that evaluate flow properties within fixed spatial domains, Lagrangian approaches track the movement and evolution of individual elements within the flow field (Esposti Ongaro et al., 2020).

The plume formulation presented by Lastic et al. (2022) and implemented in AnitoPlume (Del Gallego et al., 2026) represents plume dynamics through discrete elements with associated physical properties. Each plume element maintains state variables such as position, velocity, mass, density, and volume. During each simulation step, these variables are updated based on governing physical equations that consider forces including gravity, buoyancy, atmospheric drag, and wind effects (Lastic et al., 2022; Del Gallego et al., 2026; Folch, Costa, & Macedonio, 2016).

The structure of the Lagrangian formulation exhibits data-parallel characteristics because individual plume elements can be updated primarily using their local state information and shared simulation parameters. Consequently, operations such as force evaluation and numerical integration can be distributed across multiple processing units, provided that synchronization requirements are properly managed (Lastic et al., 2022).

However, Lagrangian particle-based methods also present limitations when applied to large-scale geophysical phenomena. The computational cost associated with representing the large number of particles required for realistic volcanic processes has limited the widespread use of fully Lagrangian approaches in volcanological simulations (Esposti Ongaro et al., 2020). These limitations motivate the consideration of alternative formulations, model refinements, or modifications to the underlying mathematical representation depending on the accuracy and computational requirements of the simulation. Consequently, the selection of governing equations and the formulation of the plume model remain important considerations in the continued development of volcanic simulation systems.

Despite these considerations, the data-parallel characteristics of the Lagrangian representation make plume simulation suitable for optimization through modern parallel processing architectures. Since the computational workload consists of continuously updating large numbers of dynamic elements, improving execution efficiency requires an architecture capable of reducing overhead and efficiently utilizing available hardware resources. Explicit graphics architectures provide mechanisms that support this objective by enabling greater control over resource management and parallel command execution.

Chapter 4

Methodology

The primary objective of this research is to migrate and improve AnitoPlume's real-time simulation framework by modernizing its rendering pipeline, optimizing hardware utilization through new, explicit graphics APIs, and refining its underlying physical simulation models.

4.1 Systems Analysis

We will investigate contemporary, explicit rendering paradigms to identify optimization opportunities within the existing legacy graphics pipeline (OpenGL). This involves studying low-level hardware abstraction techniques designed to reduce CPU overhead and improve memory bandwidth utilization. In parallel, existing computational fluid dynamics and particle-based simulation models will be evaluated to explore additional advanced algorithmic approaches to validate and potentially enhance the accuracy, scalability, and performance of the current physics implementations.

4.2 Implementation and Migration

The existing graphics architecture will be transitioned to a new, explicit graphics API. This pipeline re-engineering involves restructuring the rendering engine to utilize decoupled shading layers and unified abstractions. Concurrently, the underlying physical simulation logic, such as Lagrangian-based particle systems, will be validated and iteratively improved. Algorithmic refinements will be introduced to increase the fidelity of fluid-terrain interactions and volumetric behavior without compromising real-time performance constraints. Throughout this process, the system will undergo continuous integration and profiling. Optimization techniques will then be applied to resolve any bottlenecks in both graphics processing and physics computations.

4.3 Usability Testing

To verify that the architectural overhaul maintains or improves the interactive user experience, an exploratory usability test will be conducted with target beneficiaries:

- Setup: After screening and obtaining consent, participants will be given time to explore the interface.
- Task: Participants will use the software to recreate specific Taal Volcano eruption plumes, following a set of predefined scenarios.
- Survey: Users will complete the System Usability Scale (SUS) questionnaire to rate the software's ease of use.
- Feedback: A closing interview will gather suggestions for improving the interface and overall performance.

Table 2: Sample SUS questionnaire implemented by (Del Gallego et al., 2026) 1 — strongly disagree, 5 — strongly agree.

Code	Question
Q1	I think that I would use AnitoPlume frequently.
Q2	I found AnitoPlume unnecessarily complex.
Q3	I found AnitoPlume easy to use.
Q4	I think that I would need the support of a technical person to be able to effectively and efficiently use AnitoPlume.
Q5	I found that various features of the simulator were well integrated in AnitoPlume.
Q6	I thought there was too much inconsistency in AnitoPlume.
Q7	I would imagine that most people without prior experience in using similar software would easily learn how to use AnitoPlume.
Q8	I found AnitoPlume very cumbersome (slow or complicated and therefore inefficient) to use.
Q9	I felt very confident while using AnitoPlume.
Q10	I needed to figure out a lot of things (such as hotkeys, navigating the UI) before I could confidently use AnitoPlume.

Appendix A

Research Ethics Documents

A.1 A.1 Clearance Form

RESEARCH / PROJECT ETHICS REVIEW FORM
For Thesis / Capstone / Dissertation Proposals
(*ver.Oct2024c*)

Names of Student Researcher(s):
Adrian Rafael Bernandino
Anton Miguel Borromeo
Russell Emmanuel Galan
Reinman Geoffrey Ganayo

Research/Thesis/Capstone Title: Modernizing 3D Graphics Rendering for Interactive Taal Volcano Plume Simulation

College: College of Computer Studies

Department: Software Technology

Course: BS Computer Science major in Software Technology

Expected Duration of the Project: from: Aug 6, 2026 to Aug 6, 2027

PROTOCOL:
(*To be filled up by proponents; Use the Ethics Checklists as guides in determining areas for ethical concern/consideration. Attach and properly reference the accomplished ethics checklists and supporting documents.)*

		Reference Document (include page number)
Target participants	10 to 20 voluntary participants consisting of collegiate computer science students, graphics programmers, or software developers.	Scope and Limitations (page x-x)
Data to be used / collected	**Technical Data:** Empirical hardware performance logs including Frame Rate (FPS), CPU/GPU resource utilization percentage, and frame rendering times (ms). **Human Data:** Quantitative usability scores via the System Usability Scale (SUS) questionnaire and qualitative feedback regarding visual artifacts or rendering anomalies. No personally identifiable information (PII) such as names, student numbers, or email addresses will be tied to the responses.	Scope and Limitations (page x-x)
Brief procedure	1. **Multi-API Execution:** Run the legacy OpenGL *AnitoPlume* alongside the newly developed modern low-level explicit API implementations (such as Vulkan or DirectX 12) under identical hardware	Scope and Limitations (page x-x)

	environments to capture raw performance data. 2. **Informed Consent:** Present participants with an Informed Consent Form (ICF) outlining the evaluation's scope, hardware interactions, and data privacy measures before any testing occurs. 3. **Cross-API Usability Testing:** Participants will interact with the different API implementations of the interactive 3D plume simulator to evaluate real-time responsiveness and stability. 4. **Questionnaire Evaluation:** Participants will complete an anonymous post-test SUS questionnaire to measure and compare usability and visual consistency across the target graphics backends.	
Potential risks		
Applicable Terms of Use or Licensing Agreement of Data, Models and/or Tools	**AnitoPlume Source Code:** Used with explicit permission and under the supervision of the thesis adviser, Neil Del Gallego. **Libraries:** Eigen (Mozilla Public License v2.0) and dearIMGUI (MIT License). **Graphics Toolkits & SDKs:** Khronos Vulkan SDK, Microsoft DirectX 12 SDK, and Khronos OpenGL specifications.	Scope and Limitations (page x-x)
Steps to safeguard the participants and/or data	**Participant Safety:** Testing sessions will be kept brief (under 20 minutes) to prevent eye strain, and participants may withdraw from the usability test at any point without penalty. **Data Safety:** All performance logs and usability questionnaire responses will be fully anonymized using alphanumeric identifiers (e.g., User01, User02). Survey sheets and digital spreadsheets will be secured in a password-protected cloud storage repository accessible only to the primary student researchers, to be permanently deleted upon completion of the thesis defense	Scope and Limitations (page x-x)
Other concerns and considerations	None at this point in time	

PANEL'S ASSESSMENT/COMMENTS:

PANEL'S DECISION:

- € **Approved** (*Proponent/s may proceed with the study/data collection*)
- € **Modifications Required** (*Proponent/s should address ethical issues, follow the panel's recommendations for required modifications, and resubmit Ethics Review Form for approval before proceeding with the study or before starting with data collection*)
- € **For Full Review** (*Proponent/s should submit the proposal to the University Research Ethics Review Committee (RERC) and obtain a research ethics clearance before proceeding with the study or before starting with data collection*)

JUSTIFICATION:
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Name and Signature of Adviser/Mentor

Date: ____________________

____________________ ____________________

____________________ ____________________

Name and Signature of Panel Member **Name and Signature of Panel Member**

Date: ____________________ **Date:** ____________________

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Name and Signature of Panel Member

Date: ______________________

A.2 A.2 General Checklist

DE LA SALLE UNIVERSITY
General Research Ethics Checklist

This checklist is to ensure that the research conducted by the faculty members and students of De La Salle University is carried out according to the guiding principles outlined in the Code of Research Ethics of the University. The investigator is advised to refer to the De La Salle University Code of Research Ethics and Guide to Responsible Conduct of Research before completing this checklist. Statements pertinent to ethical issues in research should be addressed below. The checklist will help the researcher/s and advisers/readers/evaluators determine whether procedures should be undertaken during the course of the research to maintain ethical standards. The University's Guide to the Responsible Conduct of Research provides details on these appropriate procedures.

Researcher Details	
Students	**BERNANDINO**, Adrian Rafael DC. **BORROMEO**, Anton Miguel G. **GALAN**, Russell Emmanuel G. **GANAYO**, Reinman Geoffrey A.
Thesis Adviser	**DEL GALLEGO**, Neil
Department/College	Department of Software Technology / College of Computer Studies
Proposed Title of the Research	Modernizing 3D Graphics Rendering for Interactive Taal Volcano Plume Simulation
Term(s) and academic year in which research project is to be undertaken	Term 3 of AY 25-26 to Term 3 of AY 26-27

This checklist must be completed AFTER the* De La Salle University Code of Ethics *has been read and BEFORE gathering data.

Questions	Yes	No
1. Does your research involve human participants (this includes new data gathered or using pre-existing data)? If your answer is **yes**, please answer **Checklist A (Human Participants).** **Please specify if the kind of research you will be conducting falls under any of the following Human Participants sub-categories:**	✔	

1.A. Will you be conducting Action Research in an existing business, company, or school? If your answer is **yes**, please answer **Checklist F (Action Research).**		✔
1.B. Does your research involve online communities (this includes culling data from social media platforms, online forums and blogs)? If your answer is **yes**, please answer **Checklist G (Internet Research).**		✔
1.C. Does your research involve human participants who are situated in a community and may necessitate permission to acquire access to them? If your answer is **yes**, please answer **Checklist H (Community Research).**		✔
2. Will your research make use of documents which are not in the public domain and, thus, require permission for use from the custodian of such documents? **If YES, please provide certification that permission from the custodian of the data was sought and granted.**		✔
3. Will your research make use of secondary data (e.g., surveys, inventories, plans, official documents, etc.) from an institution, organization, or agency, which are not in the public domain and, thus, require permission for use from the custodian of such documents? **If YES, please provide certification that permission to use the data was sought from the institution, organization, or agency and approval was granted.**		✔
4. Does your research involve animals (non-human subjects)? If your answer is **yes**, please answer **Checklist B (Animal Subjects).**		✔
5. Does your research involve Wildlife? If your answer is **yes**, please answer **Checklist C (Wildlife)**.		✔
6. Does your research involve microorganisms that are infectious, disease causing or harmful to health? If your answer is **yes**, please answer **Checklist D (Infectious Agents).**		✔
7. Does your research involve toxic/chemicals/ substances/materials? If your answer is **yes**, please answer **Checklist E (Toxic Agents).**		✔

Research with Ethical Issues to address:

If you have a YES answer to any of the above categories, you will be required to complete a detailed checklist for that particular category. A YES answer does not mean the disapproval of your research proposal. By providing you with a more detailed checklist, we ensure that the ethical concerns are identified so these can be addressed in adherence to the University Code of Ethics.

Declaration of Conflict of Interest

[✔] 1. I do not have a conflict of interest in any form (personal, financial, proprietary, or professional) with the sponsor/grant-giving organization, the study, the co-investigators/personnel, or the site.

[ ] 2. I do have a conflict of interest, specifically:

[ ] A. I have a personal/family or professional interest in the results of the study (family members who are co-proponents or personnel in the study, membership in relevant professional associations/organizations).
Please describe the personal/family or professional interest:

[ ] B. I have proprietary interest vested in this proposal (with the intent to apply for a patent, trademark, copyright, or license)
Please describe proprietary interest:

[ ] C. I have significant financial interest vested in this proposal (remuneration that exceeds P250,000.00 each year or equity interest in the form of stock, stock options or other ownership interests).
Please describe financial interest:

A.3 A.3 Checklist A - Human Participants

DE LA SALLE UNIVERSITY

Checklist A
Research Ethics Checklist for Investigations involving Human Participants

This checklist must be completed AFTER the* De La Salle University Code of Research Ethics and Guide to Responsible Conduct of Research *has been read and BEFORE gathering data. The University Code of Research Ethics is available at http://www.dlsu.edu.ph/offices/urco/forms/URCO-Code-of-Research-Ethics_August2011.pdf

NOTE: This checklist is completed after the research proponent fills out the General Checklist Form.

Only answer this Checklist if you answered YES on question 1 of the General Checklist.

Researcher Details	
Students	**BERNANDINO**, Adrian Rafael DC. **BORROMEO**, Anton Miguel G. **GALAN**, Russell Emmanuel G. **GANAYO**, Reinman Geoffrey A.
Thesis Adviser	**DEL GALLEGO**, Neil
Department/College	Department of Software Technology / College of Computer Studies
Proposed Title of the Research	Modernizing 3D Graphics Rendering for Interactive Taal Volcano Plume Simulation
Term(s) and academic year in which research project is to be undertaken	Term 3 of AY 25-26 to Term 3 of AY 26-27

Provide a brief description of the data collection procedure to be undertaken in the research:

Data collection consists of a two-pronged approach: a technical hardware benchmarking phase and a human usability testing phase. First, the research team will execute automated, software-driven rendering loops comparing the legacy OpenGL engine against experimental low-level explicit graphics API implementations (e.g., Vulkan or DirectX 12) under identical hardware environments to capture frames per second (FPS), frame render times, and CPU/GPU utilization logs. Second, targeted human usability tests will be conducted. Voluntarily recruited computer science students or graphics developers will interact with the multi-API 3D plume simulator configurations. After completing standard interaction tasks, participants will complete an anonymous, self-administered questionnaire using the System Usability Scale (SUS) to evaluate application responsiveness, stability, and visual parity

The following should be attached to the checklist:

- A copy of the informed consent form to be used in the study.
- A copy of the instrument/tool that will be administered to the participants.
- If applicable, a copy of the letter seeking permission to collect data from participants who are under the supervision of an agency, institution, department, or office.
- If applicable, a copy of the parental consent form for participants below 18 years old.

The following items refer to important ethical considerations in the conduct of research with human participants. Provide a check for the appropriate answer to each question.

Source of data

Please check all that apply:

✔	**1. New data will be collected from human participants** ou checked this item, how will the new data be gathered? Please check all that apply. **ter answering this question, please proceed to page 3**	
	✔	**Experimental Procedures/Intervention/ Treatments**
		Focus Group
		Personal Interviews
	✔	**Self-administered Questionnaire**
		Researcher-administered Questionnaire
		Internet survey
		Observation
		Telephone survey
		Others, please specify:
	2. Pre-existing data from human participants, i.e., from a dataset **/ou checked this item, please proceed to page 7**	

options are checked (both new data and pre-existing data), **answer all of the questions in this document.**

Only answer if new data will be collected (item 1 above)	
Sampling Details	
Number of Participants/Subjects	**10 to 20 voluntary participants.**
Location where the participants will be recruited/ where subjects will be obtained?	**On-campus recruitment within DLSU computer laboratories, online developer groups, and via recommended graphics programming experts.**
How long will the data collection take place?	**2 to 4 weeks during the systems evaluation timeframe.**
Who will perform the data collection?	**The primary student proponents (Bernandino, Borromeo, Galan, and Ganayo) under adviser supervision.**
Location(s) where data collection will take place	**DLSU College of Computer Studies (CCS) laboratory environments or controlled remote developer testing environments.**
What procedures will be employed to ensure voluntary consent from participants?	**Prior to the evaluation session, participants will receive a printed or digital Informed Consent Form (ICF) explicitly stating that participation is entirely voluntary, they may exit the software at any stage, and can choose to withdraw from the evaluation without academic or personal penalty.**
Data Retention	
How long will data with participant identifiers be kept after the publication of the first paper from the project?	**0 days (No personal identifiers will be collected from the beginning; all records remain fully anonymous).**
How long will anonymized data be kept after the publication of the first paper from the project?	**2 to 3 years following final publication or degree conferral to allow for reproducibility audits, securely saved in password-protected cloud environments before permanent deletion.**
Procedure for Informed Consent	
How will informed consent be recorded? (check all that applies) Reminder: please attach informed consent that will be used in the study	[✔] Written Consent [ ] Audio-recorded Consent [✔] Online/Email recorded Consent [ ] Others, please specify:

If you will not obtain a recorded informed consent, answer the questions that follow:

Why does the waiver of informed consent not pose a threat to the welfare and rights of the participants?

Why is recording an informed consent not practical for the proposed study?

	Yes	No	Not Applicable
1. Will the research involve students who will be receiving course credits for their participation? **If YES, please attach a copy of the consent form and a summary of the debriefing process that will help participants understand how their participation in the research has provided a relevant learning experience to the crediting course.**		✔	
2. Does the study involve participants below 18 years old or those who are unable to give their informed consent? **If YES, please attach a copy of the parental consent form.**		✔	
3. Is there a possibility that the research can induce physical and/or psychological harm to the participants? Will they experience pain or some discomfort as a result from their participation in the research? **If YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.**		✔	
4. Will the participants be deliberately falsely informed or made unaware that they are being observed? Will they be misled in a way that they will possibly object to or show unease when told of the real purpose of the study?		✔	

YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.			
5. Will the research involve the discussion of, or questions on, sensitive topics (e.g. sexual activity, substance abuse, or mental health)? **If YES, please make sure that the informed consent form explicitly states that sensitive questions will be posed and that you will safeguard the anonymity of the participants and ensure confidentiality. Please attach a copy of your informed consent form and your instrument.**		✔	
	Yes	**No**	**Not Applicable**
6. Will the research involve the administration of drugs, or other substances to the participants? **YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.** **ease also attach a description of the procedure that will ensure that the participants will be brought back to their physical and psychological states prior to their participation in the research.**		✔	
7. Will biological samples (e.g. blood, saliva, urine) be obtained from the participants? **If YES, will this involve invasive procedures? Please attach a description of these procedures.**		✔	
8. Will genetic materials be obtained from the biological samples? **If YES, please attach a description of the procedures that will ensure confidentiality. Please attach the informed consent form.**		✔	

9. Will financial inducements (other than reasonable expenses, like transportation or meal allowances) be offered to the participants for their participation in their research? **If YES, the researcher(s) should be mindful of how the inducements can influence the participants' responses or behaviors during the research. Indicate the financial inducements offered to the participants:** ____		✔	
10. Is there a possibility for groups or communities to be harmed by the dissemination of the research findings? **If YES, please attach a description of procedures to ensure the anonymity and confidentiality of the research findings.**		✔	
11. Will the results of this study have a commercial value? If yes, do you intend to apply for a patent for the output of this research? Please check: ______ Yes ______ No		✔	

FOR PROPONENTS WHO WILL GATHER NEW DATA ONLY, PLEASE STOP ANSWERING.

Use of Pre-existing Data collected from Human Participants		
Indicate the dataset from which the data for the study will be sourced		
Is the data publicly available, i.e., the access to which does not necessitate an approval process?		Yes Please indicate where the dataset is available:
		No Please indicate/attach the approval authority for access:
		Yes

Was the original dataset originally collected for the present study's purpose?		**Please attach the Consent Form used in the original study.**
		No **Please attach the Information Collection Statement (i.e., the statement given to informants providing them with the rationale for the collection of specific information).**
Does the original data set contain sensitive data, that is information that an individual would not likely want to be disclosed publicly, e.g., data on sexual activities, substance use?		Yes **Please describe the type of sensitive data to be used in the present research:**
		No
Does the original dataset have personal identifiers?		No *(This means that neither the researcher nor the participant provided any personal identifiers)*
		Yes, specifically: ______ Direct *(i.e., the participant provided personal details like name and address)* _____ Indirect (*i.e., the participant was given a respondent code to make the participant identifiable)*
Will new data be collected and analyzed along with data from the existing dataset?		Yes **Please answer questions on page 3-5.**
		No

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